

Page replacement algorithm 241401084

1)FIFO

```
#include <stdio.h>

int main()
{
    int i,j,n,a[50],frame[10],no,k,avail,count=0;

    printf("Enter number of pages: ");
    scanf("%d",&n);

    printf("Enter the page numbers:\n");
    for(i=0;i<n;i++)
        scanf("%d",&a[i]);

    printf("Enter number of frames: ");
    scanf("%d",&no);

    for(i=0;i<no;i++)
        frame[i]=-1;

    j=0;

    printf("\nRef\tFrames\n");

    for(i=0;i<n;i++)
    {
        printf("%d\t",a[i]);
        avail=0;

        for(k=0;k<no;k++)
        {
            if(frame[k]==a[i])
                avail=1;
        }

        if(avail==0)
        {
            frame[j]=a[i];
            j=(j+1)%no;
            count++;

            for(k=0;k<no;k++)
            {
                if(frame[k]!=-1)
                    printf("%d ",frame[k]);
                else
                    printf("- ");
            }

            printf("\n");
        }
        else
        {
            printf("Hit");
        }
    }

    printf("\n");

    printf("\nTotal Page Faults: %d\n",count);

    return 0;
}
```

Output:-

```

[user123@localhost bankers]$ cd page
[user123@localhost page]$ vi fifo.c
[user123@localhost page]$ gcc fifo.c
[user123@localhost page]$ ./a.out
Enter number of pages: 7
Enter the page numbers:
1 3 0 3 5 6 3
Enter number of frames: 3

Ref      Frames
1         1 - -
3         1 3 -
0         1 3 0
3         Hit
5         5 3 0
6         5 6 0
3         5 6 3

Total Page Faults: 6
[user123@localhost page]$

```

2)LRU

```

#include <stdio.h>
int findLRU(int time[], int n)
{
    int i, minimum = time[0], pos = 0;
    for(i = 1; i < n; i++)
    {
        if(time[i] < minimum)
        {
            minimum = time[i];
            pos = i;
        }
    }
    return pos;
}
int main()
{
    int no_of_frames, no_of_pages, frames[10], pages[30], counter=0, time[10];
    int flag1, flag2, i, j, pos, faults=0;
    printf("Enter number of frames: ");
    scanf("%d", &no_of_frames);
    printf("Enter number of pages: ");
    scanf("%d", &no_of_pages);
    printf("Enter reference string: ");
    for(i=0; i<no_of_pages; i++)
        scanf("%d", &pages[i]);
    for(i=0; i<no_of_frames; i++)
        frames[i]=-1;
    printf("\nRef\tFrames\n");
    for(i=0; i<no_of_pages; i++)
    {
        flag1=flag2=0;
        for(j=0; j<no_of_frames; j++)
        {
            if(frames[j]==pages[i])
            {
                counter++;
                time[j]=counter;
                flag1=flag2=1;
                break;
            }
        }
        if(flag1==0)
        {
            for(j=0; j<no_of_frames; j++)
            {

```

```

        if(frames[j]==pages[i])
        {
            counter++;
            time[j]=counter;
            flag1=flag2=1;
            break;
        }
    }
    if(flag1==0)
    {
        for(j=0;j<no_of_frames;j++)
        {
            if(frames[j]==-1)
            {
                counter++;
                faults++;
                frames[j]=pages[i];
                time[j]=counter;
                flag2=1;
                break;
            }
        }
    }
    if(flag2==0)
    {
        pos=findLRU(time,no_of_frames);
        counter++;
        faults++;
        frames[pos]=pages[i];
        time[pos]=counter;
    }
    printf("%d\t",pages[i]);
    for(j=0;j<no_of_frames;j++)
    {
        if(frames[j]!=-1)
            printf("%d ",frames[j]);
        else
            printf("- ");
    }
    printf("\n");
}
printf("\nTotal Page Faults: %d\n",faults);
return 0;
}
-- INSERT --

```

output:-

```

[user123@localhost page]$ gcc lru.c
[user123@localhost page]$ ./a.out
Enter number of frames: 3
Enter number of pages: 7
Enter reference string: 7 0 1 2 0 3 0

Ref      Frames
7         7 - -
0         7 0 -
1         7 0 1
2         2 0 1
0         2 0 1
3         2 0 3
0         2 0 3

Total Page Faults: 5
[user123@localhost page]$

```